

FIREFLY FRAMES

Fabrication drawings for the two block frames, Rib and Hoop, and the parts they share

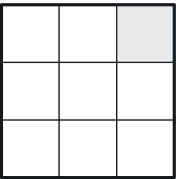
SHEETS			
SHEET	TITLE	PART NO.	SET
G-1	Drawing set index and general notes		Index
R-0	Rib frame assembly	FF-R00	Rib
R-1	Rib side walls, right and left	FF-R01, FF-R02	Rib
R-2	Rib side wall, flat pattern	FF-R01, FF-R02	Rib
R-3	Rib top and bottom walls	FF-R03, FF-R04	Rib
R-4	Rib seam rail	FF-R05	Rib
R-5	Rib hub hanger	FF-R06	Rib
R-6	Rib corner and tab details	FF-R01 to FF-R04	Rib
H-0	Hoop frame assembly	FF-H00	Hoop
H-1	Hoop strip: profile, corners and weld	FF-H01	Hoop
H-2	Hoop strip, flat pattern	FF-H01	Hoop
H-3	Hoop strip, punch details	FF-H01	Hoop
H-4	Hoop seam rail	FF-H02	Hoop
H-5	Hoop hub mount	FF-H03, FF-H04, FF-H05	Hoop
C-1	Gravity lock, panel to frame	J groove	Common
C-2	Block joints	Joint features	Common
C-3	Mounting: hole grid, angle and tube clips	FF-C05	Common
C-4	Joint bolts and hardware	ISO 4162, CLS-M8-1, DIN 929	Common
C-5	Logo strip and finish	Side walls, all blocks	Common

THE FOUR BLOCKS, NAMED HEIGHT x WIDTH IN FEET

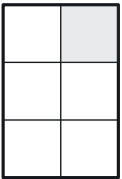
BLOCK	OUTSIDE W x H x D	PANELS	JOINT BOLTS	RIB FRAME, kg	HOOP FRAME, kg
3x3	914.4 x 914.4 x 152.4	9	8	5.4	12.7
3x2	609.6 x 914.4 x 152.4	6	8	4.1	9.4
2x3	914.4 x 609.6 x 152.4	6	8	4.2	10.1
2x2	609.6 x 609.6 x 152.4	4	8	3.2	7.4

- The 3x2 is 3 high and 2 wide, a part of its own, never a 2x3 turned on end. Every block hangs one way only.
- Frame masses are approximate, bare frame before powder coat, without panels or hub.

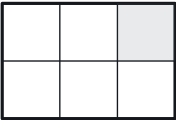
FROM THE FRONT, 1:40. THE HUB SITS BEHIND THE SHADED PANEL.



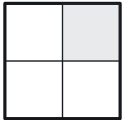
3x3



3x2



2x3



2x2

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

G-1

GENERAL NOTES

- Dimensions in millimeters; inches in [] for reference. Third angle projection. Do not scale the drawings.
- Tolerances unless stated: laser and punched features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius equal to the sheet thickness (K 0.40). Break sharp edges 0.2 and remove burrs.
- Rib: 5052-H32 aluminum, 1.6 walls and 2.0 rails and hanger, folded and riveted with flush blind rivets. Hoop: S280GD+ZM310 steel 1.2, punched, roll formed, bent and closed with one weld; 2.0 DX51D+Z275 for the hub mount parts.
- Finish: matte black polyester powder RAL 9005, 60 to 80 μm , over the pretreatment on each sheet. Mask every nut thread and every $\varnothing 8.4$ joint hole.
- Logo strip: three Cirrus logos to every foot up both side walls of every block, centers 50.8 [2] up and then every 101.6 [4], 17 back from the front edge, printed in gloss over the matte coat (C-5). The front 32 of each side wall carries no hole, rivet, fastener, clinch, weld or tool mark.
- Every block joins every other: four M8 bolts on every side, on two lines 38.1 and 111.1 back from the front edge, in the middle of the first foot and the last (C-2). Rib and Hoop blocks join each other and share the panels, hubs, mounting steel and hardware.
- Front service: a panel lifts 12.7 and comes out the front, then the hub comes out through the same opening. Nothing that holds the hub is fixed to a side wall and no cable passes through a frame wall.
- Part numbers: FF-R for Rib, FF-H for Hoop, FF-C for parts both frames use. A suffix of two digits gives the block, height then width (FF-R00-32 is the 3x2 Rib frame); one digit gives a length in feet.
- These sheets are preliminary, for quotation and engineering review. Quote each frame for each block size, with any tooling as its own line.

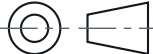
RIB, HOW IT IS MADE

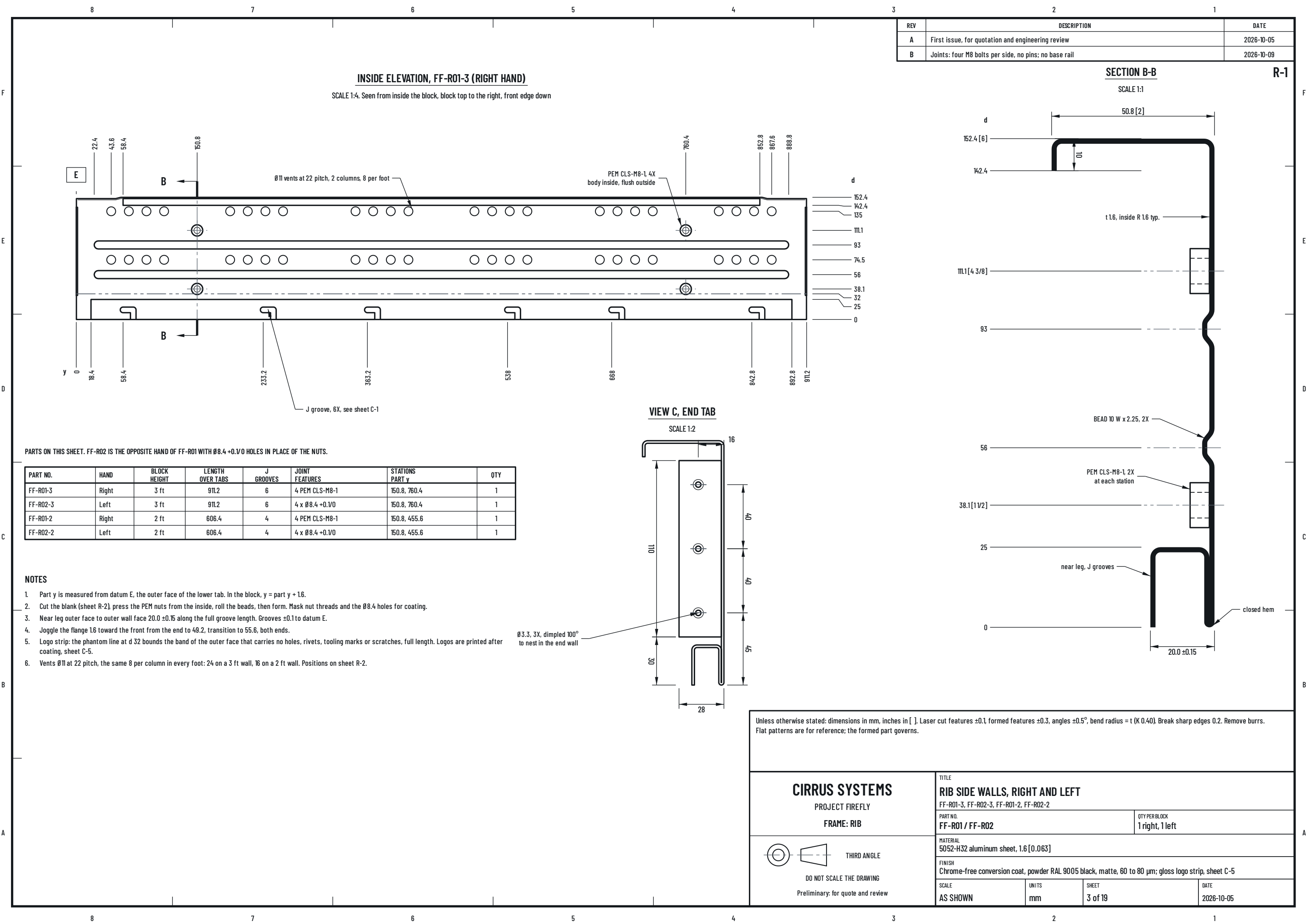
- Laser cut or punch the flat blanks: walls, seam rails and hanger (R-2 to R-5).
- Press the PEM nuts into the flat blanks and dimple the rivet holes.
- Form on a press brake: side walls with their beads, top and bottom walls, rails, hanger.
- Conversion coat and powder coat every part matte black, masking nut threads and joint holes. Pad print the gloss logo strip on the side walls (C-5).
- Rivet on a fixture: side wall tabs inside the top and bottom walls, the corners through the flanges, the seam rails, then the hanger. Every rivet flush.
- Check nut to hole positions and the diagonals.

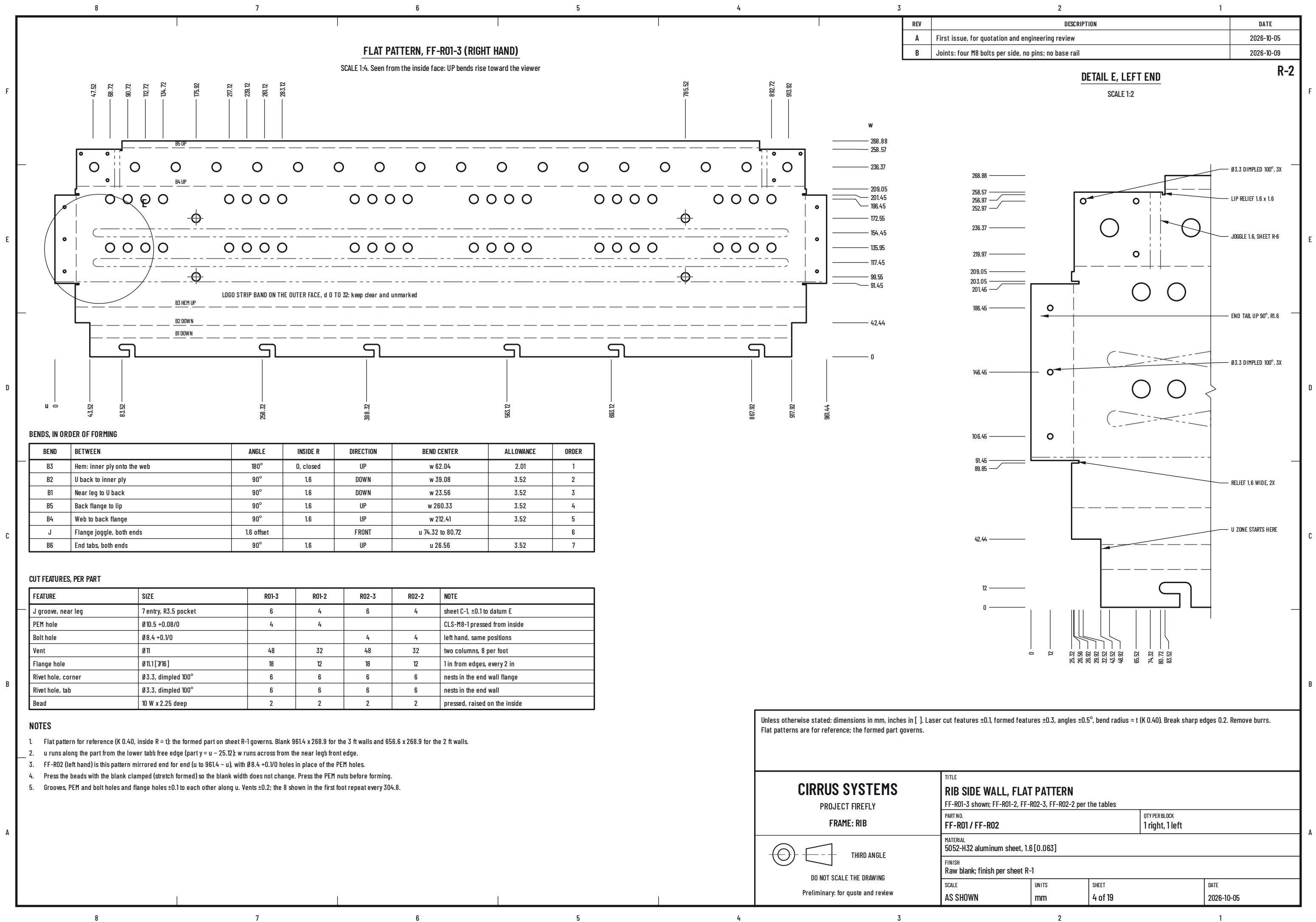
HOOP, HOW IT IS MADE

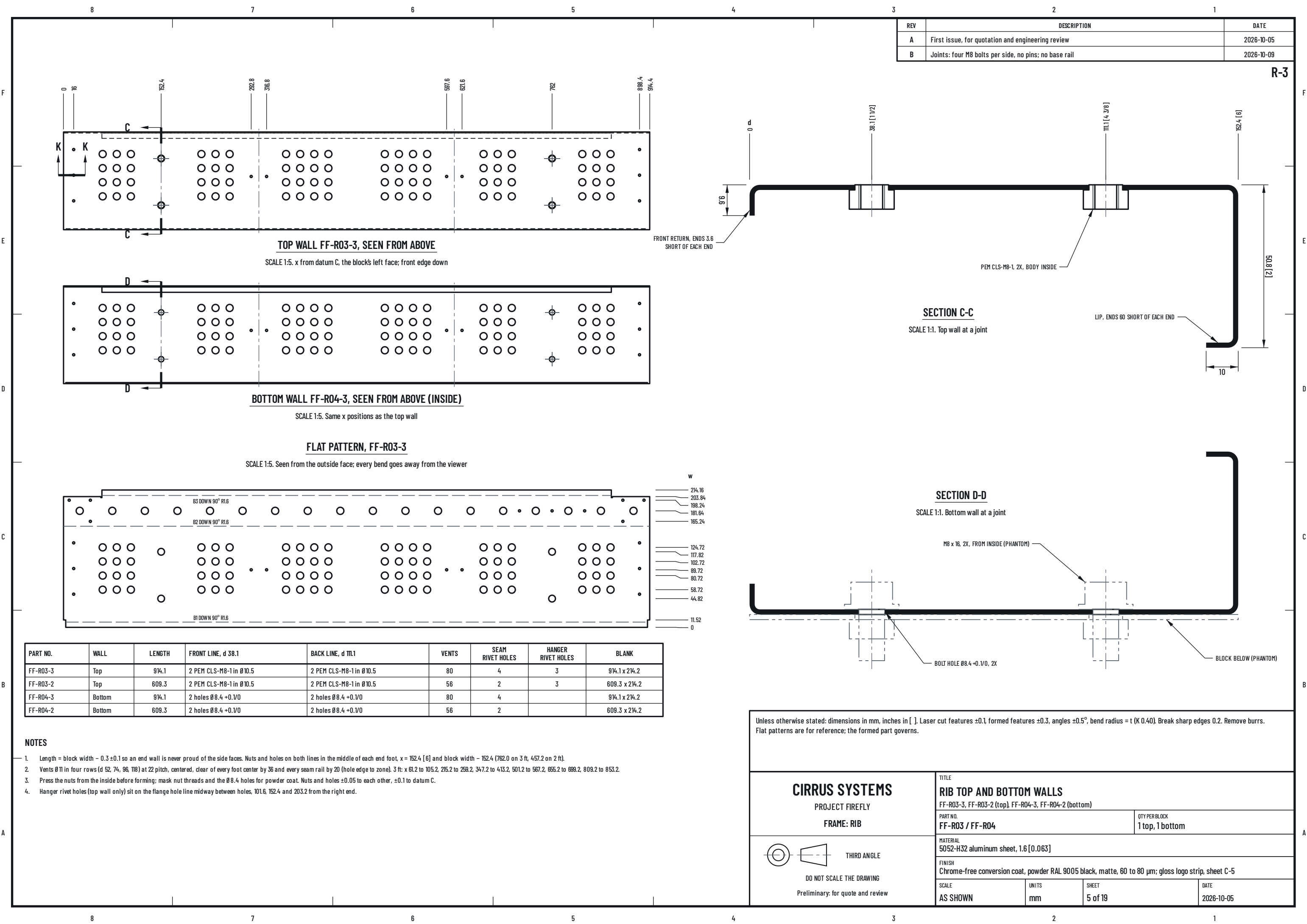
- Punch the coil in register: holes, J grooves, vents and corner notches (H-2, H-3).
- Roll form the profile (H-1) and cut to length.
- Bend the four R12 corners and close the ring with one butt weld.
- Projection weld the M8 nuts from inside.
- Clinch in the seam rails, the hinge shelf with its pins, and the latch strap.
- Mask, clean, pretreat and powder coat matte black. Pad print the gloss logo strip (C-5), then fit the latch.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY RIB AND HOOP THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE DRAWING SET INDEX Sheets, the four blocks and the general notes			
	PART NO. FIREFLY FRAMES		QTY PER BLOCK	
	MATERIAL Per sheet			
	FINISH Per sheet			
	SCALE AS SHOWN	UNITS mm	SHEET 1 of 19	DATE 2026-10-05







REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

R-3

FRONT RETURN, ENDS 3.6
SHORT OF EACH END

PEM CLS-M8-1, 2X, BODY INSIDE

LIP, ENDS 60 SHORT OF EACH END

10

50.8 [2]

SECTION D-D

SCALE 1:1. Bottom wall at a joint

M8 x 16, 2X, FROM INSIDE (PHANTOM)

BOLT HOLE Ø8.4 +0.1/0, 2X

BLOCK BELOW (PHANTOM)

PART NO.	WALL	LENGTH	FRONT LINE, d 38.1	BACK LINE, d 111.1	VENTS	SEAM RIVET HOLES	HANGER RIVET HOLES	BLANK
FF-R03-3	Top	914.1	2 PEM CLS-M8-1 in Ø10.5	2 PEM CLS-M8-1 in Ø10.5	80	4	3	914.1 x 214.2
FF-R03-2	Top	609.3	2 PEM CLS-M8-1 in Ø10.5	2 PEM CLS-M8-1 in Ø10.5	56	2	3	609.3 x 214.2
FF-R04-3	Bottom	914.1	2 holes Ø8.4 +0.1/0	2 holes Ø8.4 +0.1/0	80	4		914.1 x 214.2
FF-R04-2	Bottom	609.3	2 holes Ø8.4 +0.1/0	2 holes Ø8.4 +0.1/0	56	2		609.3 x 214.2

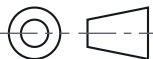
NOTES

- Length = block width - 0.3±0.1 so an end wall is never proud of the side faces. Nuts and holes on both lines in the middle of each end foot, x = 152.4 [6] and block width - 152.4 (762.0 on 3 ft, 457.2 on 2 ft).
- Vents Ø11 in four rows (d 52, 74, 96, 118) at 22 pitch, centered, clear of every foot center by 36 and every seam rail by 20 (hole edge to zone). 3 ft: x 61.2 to 105.2, 215.2 to 258.2, 347.2 to 413.2, 501.2 to 567.2, 655.2 to 698.2, 809.2 to 853.2.
- Press the nuts from the inside before forming; mask nut threads and the Ø8.4 holes for powder coat. Nuts and holes ±0.05 to each other, ±0.1 to datum C.
- Hanger rivet holes (top wall only) sit on the flange hole line midway between holes, 101.6, 152.4 and 203.2 from the right end.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: RIB



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

RIB TOP AND BOTTOM WALLS

FF-R03-3, FF-R03-2 (top), FF-R04-3, FF-R04-2 (bottom)

PART NO.

FF-R03 / FF-R04

QTY PER BLOCK

1 top, 1 bottom

MATERIAL

5052-H32 aluminum sheet, 1.6 [0.063]

FINISH

Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-5

SCALE

AS SHOWN

UNITS

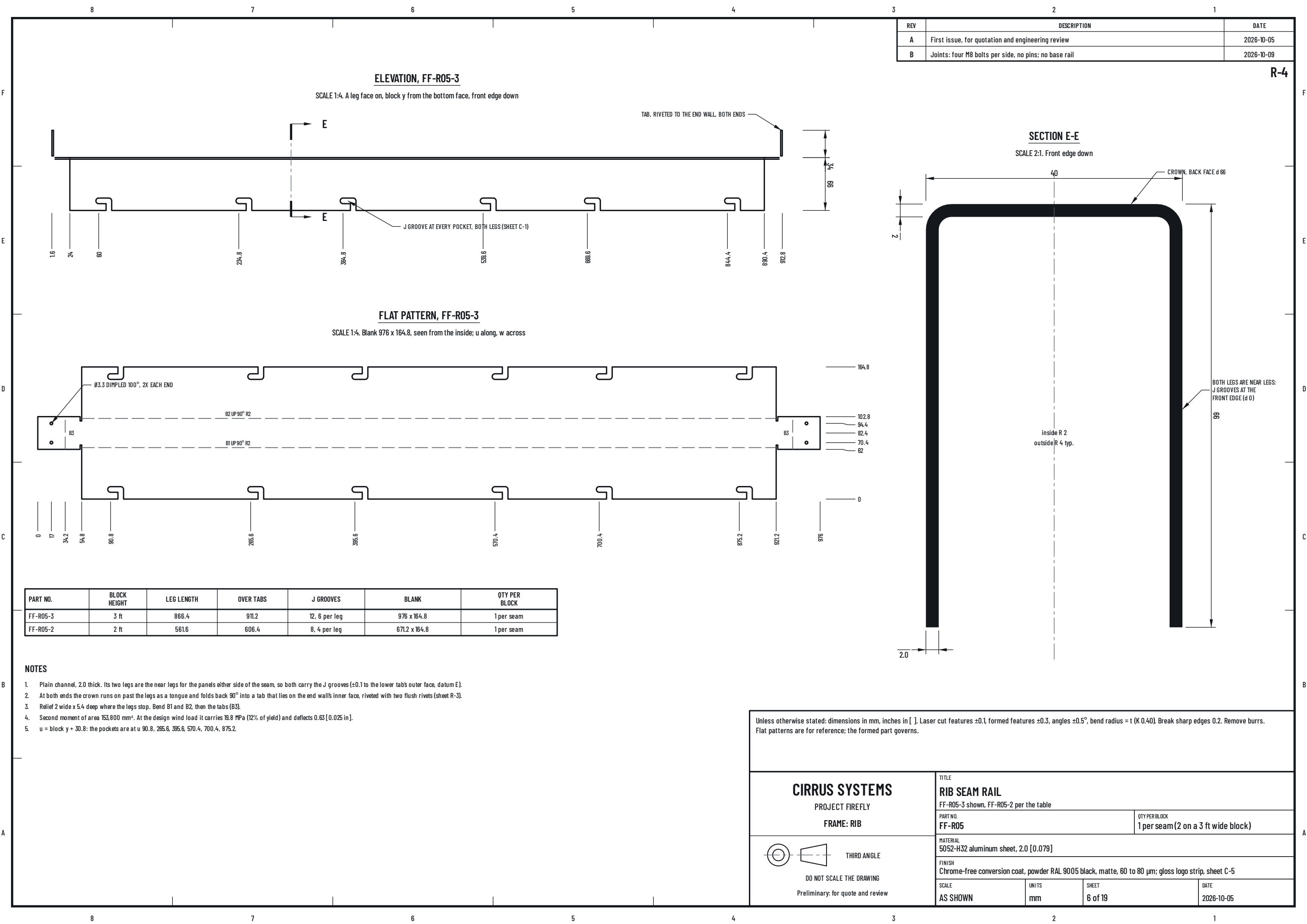
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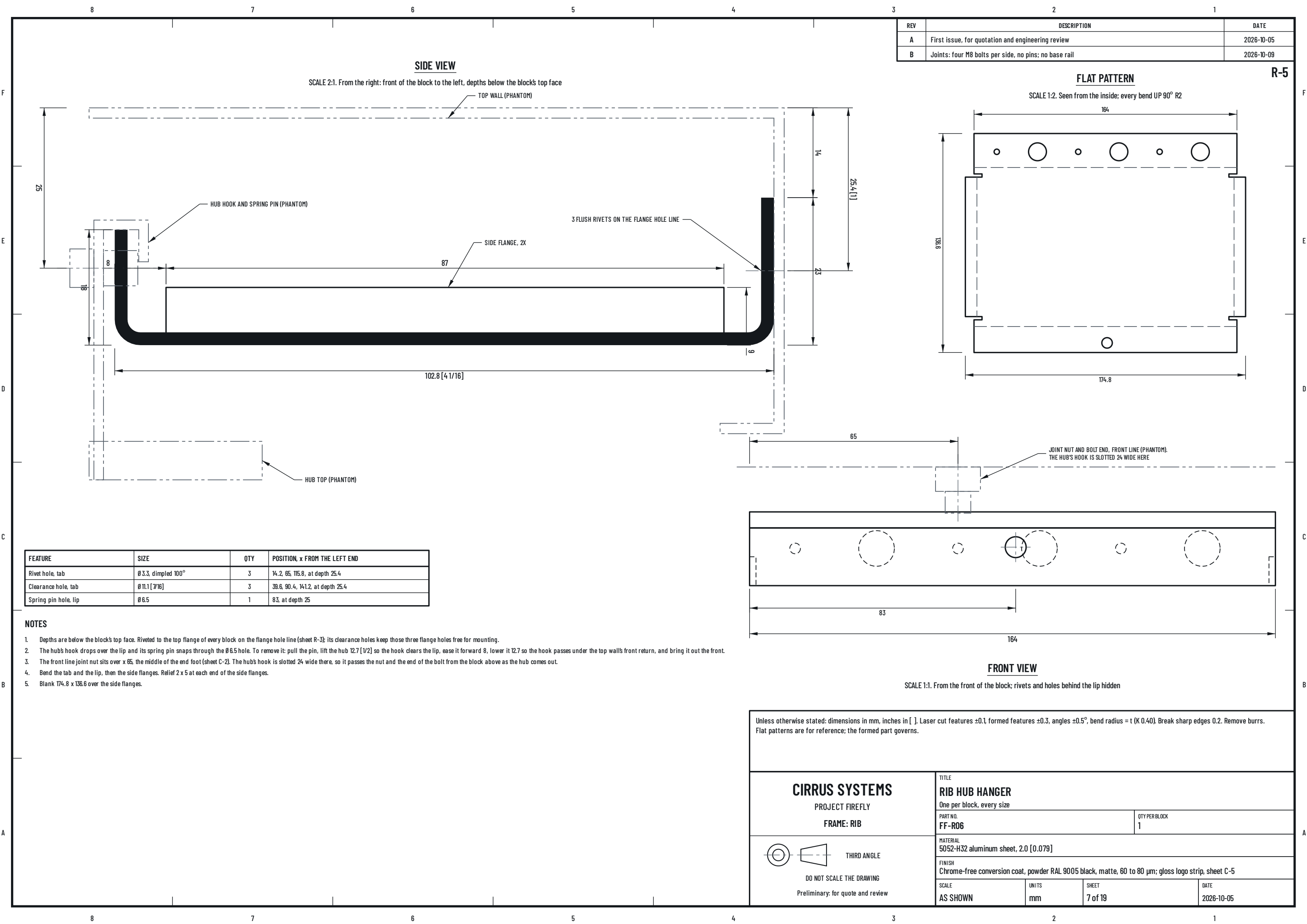
SHEET

5 of 19

DATE

2026-10-05

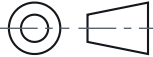


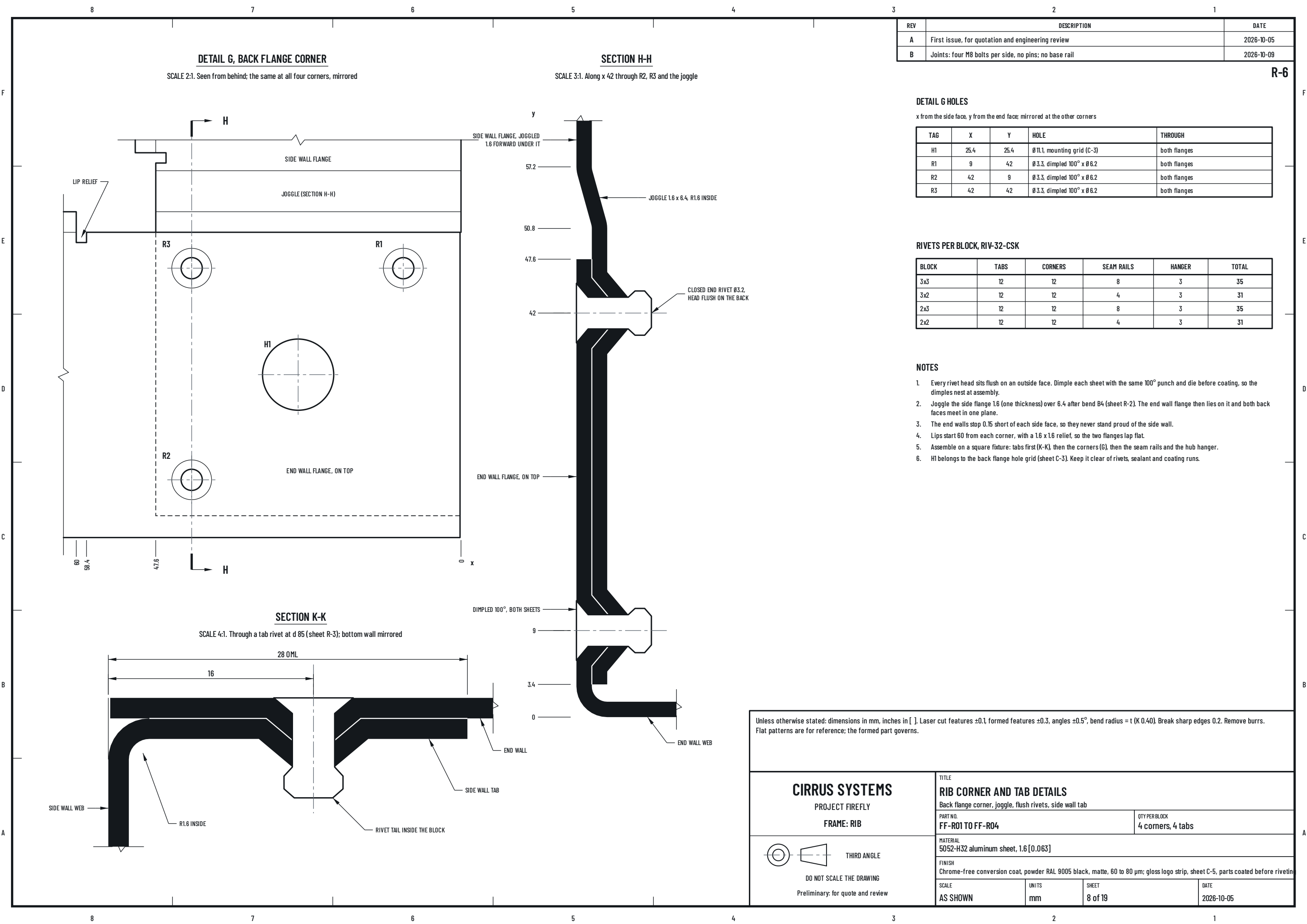


FEATURE	SIZE	QTY	POSITION, x FROM THE LEFT END
Rivet hole, tab	Ø 3.3, dimpled 100°	3	14.2, 65, 115.8, at depth 25.4
Clearance hole, tab	Ø 11.1 [7/16]	3	39.6, 90.4, 141.2, at depth 25.4
Spring pin hole, lip	Ø 6.5	1	83, at depth 25

- NOTES**
- Depths are below the block's top face. Riveted to the top flange of every block on the flange hole line (sheet R-3); its clearance holes keep those three flange holes free for mounting.
 - The hub's hook drops over the lip and its spring pin snaps through the Ø 6.5 hole. To remove it: pull the pin, lift the hub 12.7 [1/2] so the hook clears the lip, ease it forward 8, lower it 12.7 so the hook passes under the top wall's front return, and bring it out the front.
 - The front line joint nut sits over x 65, the middle of the end foot (sheet C-2). The hub's hook is slotted 24 wide there, so it passes the nut and the end of the bolt from the block above as the hub comes out.
 - Bend the tab and the lip, then the side flanges. Relief 2 x 5 at each end of the side flanges.
 - Blank 174.8 x 136.6 over the side flanges.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB  THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB HUB HANGER One per block, every size			
	PART NO. FF-R06		QTY PER BLOCK 1	
	MATERIAL 5052-H32 aluminum sheet, 2.0 [0.079]			
	FINISH Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-5			
	SCALE AS SHOWN	UNITS mm	SHEET 7 of 19	DATE 2026-10-05



REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
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R-6

DETAIL G HOLES

x from the side face, y from the end face; mirrored at the other corners

TAG	X	Y	HOLE	THROUGH
H1	25.4	25.4	Ø 11.1, mounting grid (C-3)	both flanges
R1	9	42	Ø 3.3, dimpled 100° x Ø 6.2	both flanges
R2	42	9	Ø 3.3, dimpled 100° x Ø 6.2	both flanges
R3	42	42	Ø 3.3, dimpled 100° x Ø 6.2	both flanges

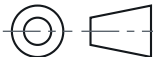
RIVETS PER BLOCK, RIV-32-CSK

BLOCK	TABS	CORNERS	SEAM RAILS	HANGER	TOTAL
3x3	12	12	8	3	35
3x2	12	12	4	3	31
2x3	12	12	8	3	35
2x2	12	12	4	3	31

NOTES

- Every rivet head sits flush on an outside face. Dimple each sheet with the same 100° punch and die before coating, so the dimples nest at assembly.
- Joggle the side flange 1.6 (one thickness) over 6.4 after bend B4 (sheet R-2). The end wall flange then lies on it and both back faces meet in one plane.
- The end walls stop 0.15 short of each side face, so they never stand proud of the side wall.
- Lips start 60 from each corner, with a 1.6 x 1.6 relief, so the two flanges lap flat.
- Assemble on a square fixture: tabs first (K-K), then the corners (G), then the seam rails and the hub hanger.
- H1 belongs to the back flange hole grid (sheet C-3). Keep it clear of rivets, sealant and coating runs.

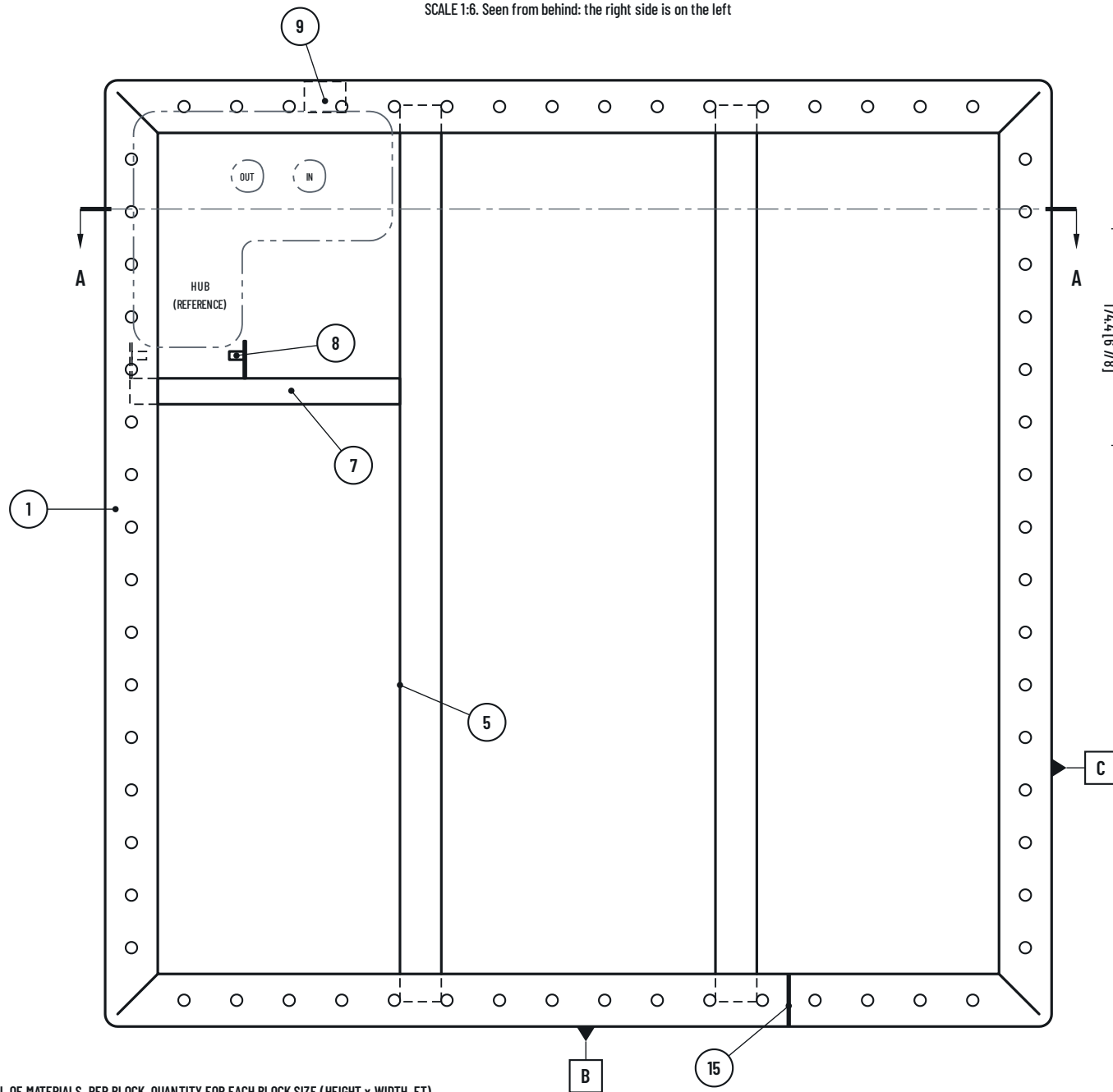
Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB CORNER AND TAB DETAILS Back flange corner, joggle, flush rivets, side wall tab			
	PART NO. FF-R01 TO FF-R04		QTY PER BLOCK 4 corners, 4 tabs	
	MATERIAL 5052-H32 aluminum sheet, 1.6 [0.063]			
	FINISH Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-5, parts coated before riveting			
	SCALE AS SHOWN	UNITS mm	SHEET 8 of 19	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

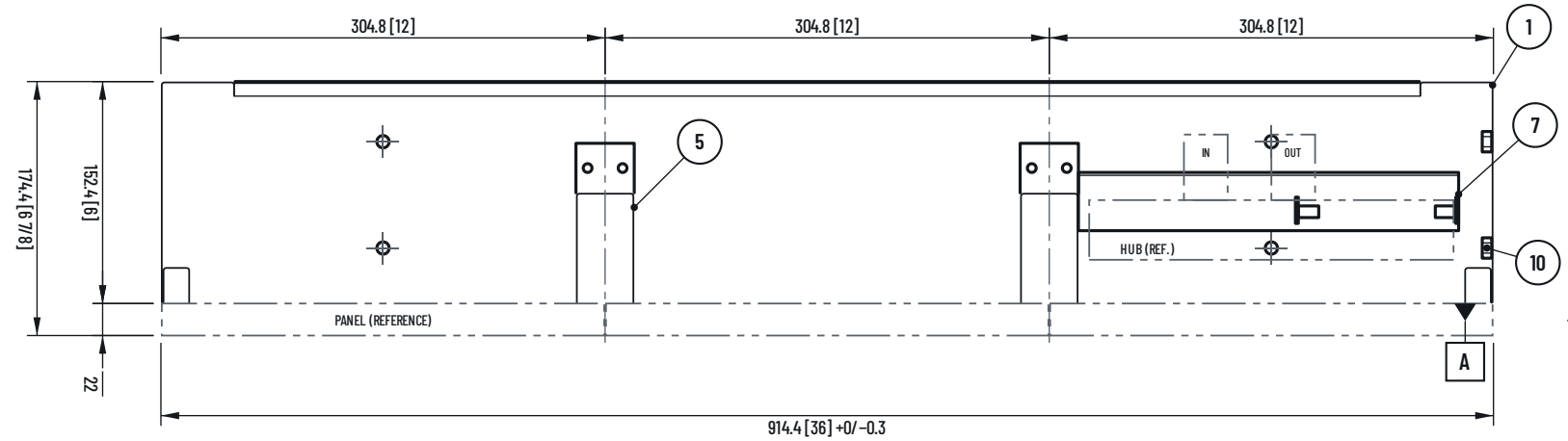
REAR VIEW, 3x3 BLOCK

SCALE 1:6. Seen from behind: the right side is on the left



SECTION A-A

SCALE 1:5. Looking down at 790 above the bottom face, front edge down. Vents and flange holes not shown



ASSEMBLY NOTES

1. Fixture on datums A (front plane), B (bottom face) and C (left face, seen from the front). Outside size $+0/-0.3$ each way, diagonals equal within 1.0, back flange flat within 0.5.
2. Order: close the ring with its one weld (sheet H-1), projection weld the nuts, rivet the hinge pins into the shelf, clinch in the seam rails, hinge shelf and latch strap, mask the hinge pins, every nut thread and the Ø8.4 joint holes, then clean, pretreat and powder coat the whole frame matte black. Pad print the gloss logo strip up both sides (sheet C-5). Fit the latch and its screws last.
3. Nothing is fastened to a side wall's outer face: it carries the logo strip and the side joints. Clinch dishes sit on the top, bottom and back faces only, never proud.
4. Joint nuts and bolt holes, four on every side: sheet C-2. Back flange holes: sheet C-3 (no corner holes on Hoop). Gravity lock grooves: sheet C-1. Hub mount: sheet H-5.
5. The hub is its own assembly, shown for reference. Its two hooks hang on item 8 and item 12 latches its top.
6. Item 14 ships loose: one bolt for each Ø8.4 hole in the bottom wall and in the left side wall (seen from the front), eight in all. Each bolts into the next block's weld nut (sheet C-2).
7. Blocks are named height x width in feet: 3x3, 3x2 (3 high, 2 wide), 2x3 (2 high, 3 wide) and 2x2.

EACH BLOCK SIZE

BLOCK	ASSEMBLY	OUTSIDE W x H x D	SEAM RAILS	STRIP, kg	FRAME, kg (APPROX.)
3x3	FF-H00-33	914.4 x 914.4 x 152.4	2	8.7	12.7
3x2	FF-H00-32	609.6 x 914.4 x 152.4	1	7.2	9.4
2x3	FF-H00-23	914.4 x 609.6 x 152.4	2	7.2	10.1
2x2	FF-H00-22	609.6 x 609.6 x 152.4	1	5.7	7.4

1. Frame mass includes the seam rails, shelf, strap, nuts, hinge pins and latch, before powder coat; the panels and hub are not included.

BILL OF MATERIALS, PER BLOCK. QUANTITY FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT).

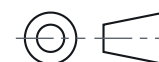
ITEM	PART NO.	DESCRIPTION	MATERIAL	3x3	3x2	2x3	2x2
1	FF-H01-33	Hoop strip for a 3x3, punched, formed, bent and welded	S280GD+ZM310, 1.2	1			
2	FF-H01-32	Hoop strip for a 3x2	S280GD+ZM310, 1.2		1		
3	FF-H01-23	Hoop strip for a 2x3	S280GD+ZM310, 1.2			1	
4	FF-H01-22	Hoop strip for a 2x2	S280GD+ZM310, 1.2				1
5	FF-H02-3	Seam rail, channel, 3 ft, clinched (sheet H-4)	S280GD+ZM310, 1.2	2	1		
6	FF-H02-2	Seam rail, channel, 2 ft, clinched	S280GD+ZM310, 1.2			2	1
7	FF-H03	Hinge shelf, clinched to the seam rail (sheet H-5)	DX51D+Z275, 2.0	1	1	1	1
8	FF-H04	Hinge pin Ø8, orbital riveted into item 7	303 stainless	2	2	2	2
9	FF-H05	Latch strap, clinched to the top flange	DX51D+Z275, 2.0	1	1	1	1
10	DIN 929	Hex weld nut M8, 4 in the top wall, 4 in the right side wall	Steel, plain	8	8	8	8
11	CLS-M3-1	PEM self clinching nut M3, in item 9	Stainless	2	2	2	2
12	LATCH	Spring bolt latch, push to release (sheet H-5)	Purchased	1	1	1	1
13	ISO 7045	Pan screw M3 x 16, latch to strap	A2 stainless	2	2	2	2
14	ISO 4162	Hex flange bolt M8 x 16, 8.8, ships loose for the joints (sheet C-2)	Steel, zinc flake	8	8	8	8
15	WELD	Square butt, one per ring, x 254 on the bottom (sheet H-1)		1	1	1	1

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: HOOP



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

	TITLE
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HOOP FRAME ASSEMBLY

3x3 shown, every block size in the bill of materials

PART NO.

FF-H00-33 / -32 / -23 / -22

QTY PER BLOCK

1

	MATERIAL
--	----------

See bill of materials

FINISH

Zirconium pretreat, powder RAL 9005 black, matte, 60 to 80 μm ; gloss logo strip, sheet C-5, after welding and clinching

SCALE

AS SHOWN

	UNITS
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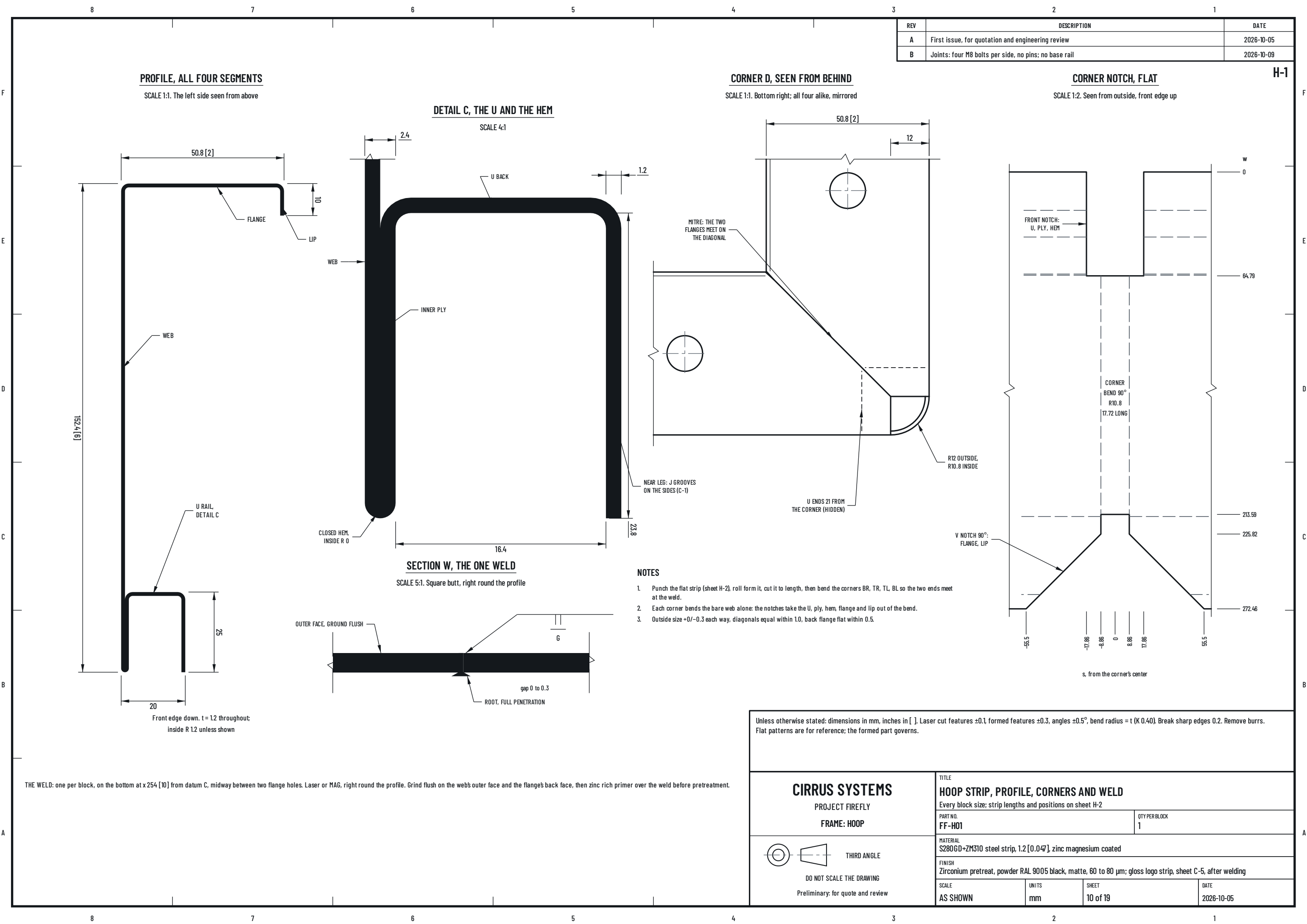
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	SHEET
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9 of 19

DATE _____

2026-10-05



REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

PROFILE, ALL FOUR SEGMENTS

SCALE 1:1. The left side seen from above

DETAIL C, THE U AND THE HEM

SCALE 4:1

CORNER D, SEEN FROM BEHIND

SCALE 1:1. Bottom right; all four alike, mirrored

CORNER NOTCH, FLAT

SCALE 1:2. Seen from outside, front edge up

H-1

NOTES

- Punch the flat strip (sheet H-2), roll form it, cut it to length, then bend the corners BR, TR, TL, BL so the two ends meet at the weld.
- Each corner bends the bare web alone: the notches take the U, ply, hem, flange and lip out of the bend.
- Outside size +0/-0.3 each way, diagonals equal within 1.0, back flange flat within 0.5.

SECTION W, THE ONE WELD

SCALE 5:1. Square butt, right round the profile

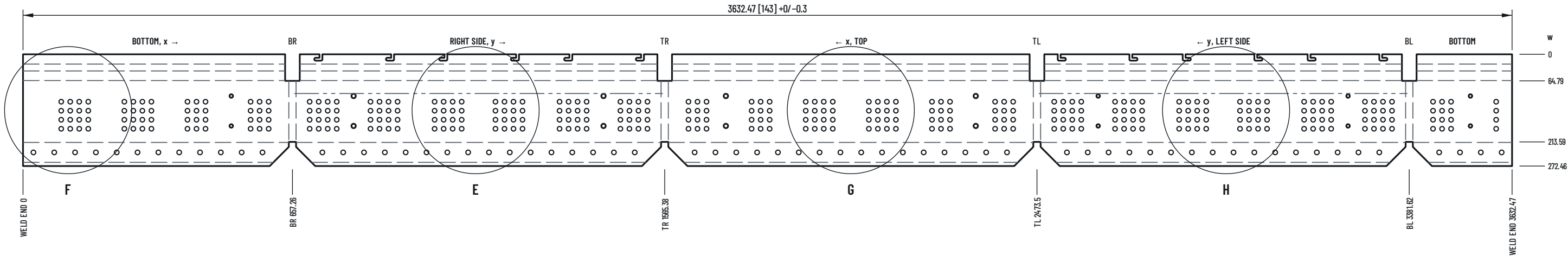
THE WELD: one per block, on the bottom at x 254 [10] from datum C, midway between two flange holes. Laser or MAG, right round the profile. Grind flush on the web's outer face and the flange's back face, then zinc rich primer over the weld before pretreatment.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.			
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP		TITLE HOOP STRIP, PROFILE, CORNERS AND WELD Every block size; strip lengths and positions on sheet H-2	
		PART NO. FF-H01	QTY PER BLOCK 1
MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
FINISH Zirconium pretreat, powder RAL 9005 black, matte, 60 to 80 μm ; gloss logo strip, sheet C-5, after welding			
SCALE AS SHOWN	UNITS mm	SHEET 10 of 19	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
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FLAT STRIP, FF-H01-33 (3x3 BLOCK)

SCALE 1:10. Seen from outside, front edge up, s from the weld end to the right



Details E to H on sheet H-3; F shows both ends butted at the weld. Corner bends run 8.86 either side of each center.

STRIP FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT)

PART NO.	BLOCK	OUTSIDE W x H	LENGTH L	s, BR	s, TR	s, TL	s, BL	MASS, kg
FF-H01-33	3x3	914.4 x 914.4	3632.47	657.26	1565.38	2473.5	3381.62	8.7
FF-H01-32	3x2	609.6 x 914.4	3022.87	352.46	1260.58	1863.9	2772.02	7.2
FF-H01-23	2x3	914.4 x 609.6	3022.87	657.26	1260.58	2168.7	2772.02	7.2
FF-H01-22	2x2	609.6 x 609.6	2413.27	352.46	955.78	1559.1	2162.42	5.7

CUT FEATURES AND HARDWARE PER STRIP

PART NO.	JOINT HOLES	VENTS	FLANGE HOLES	J GROOVES	CORNER NOTCHES	WELD NUTS DIN 929
FF-H01-33	16	344	64	12	8	8
FF-H01-32	16	296	52	12	8	8
FF-H01-23	16	280	52	8	8	8
FF-H01-22	16	232	40	8	8	8

EVERY CUT FEATURE, ALONG ITS SEGMENT IN BLOCK COORDINATES

FEATURE	SEGMENT	ACROSS, w	SIZE	ALONG, 3 FT SEGMENT	ALONG, 2 FT SEGMENT
Bolt hole	Bottom	101.69 (d 38.1), 174.69	Ø8.4 +0.1/0	x 152.4, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Top	101.69 (d 38.1), 174.69	Ø10.6 +0.1/0	x 152.4, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Right side	101.69 (d 38.1), 174.69	Ø10.6 +0.1/0	y 152.4, 762	y 152.4, 457.2
Bolt hole	Left side	101.69 (d 38.1), 174.69	Ø8.4 +0.1/0	y 152.4, 762	y 152.4, 457.2
J groove, pocket	Both sides	0 to 15.5	7 wide, R3.5	y 60, 234.8, 364.8, 539.6, 669.6, 844.4	y 60, 234.8, 364.8, 539.6
Vent, 4 rows	Bottom	115.59, 137.59, 159.59, 181.59	Ø11, 22 pitch	x 61.2 (3), 215.2, 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (2), 337.8 (4), 513.8 (3)
Vent, 4 rows	Top	the same	Ø11, 22 pitch	x 61.2 (3), 215.2 (3), 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (4), 337.8 (4), 513.8 (3)
Vent, 4 rows	Both sides	the same	Ø11, 22 pitch	y 45.2 (4), 193.6 (4), 350 (4), 498.4 (4), 654.8 (4), 803.2 (4)	y 45.2 (4), 193.6 (4), 350 (4), 498.4 (4)
Flange hole	All four	239.22 (q 25.4)	Ø11.1, 50.8 pitch	from 76.2 (16)	from 76.2 (10)

MANUFACTURING ROUTE, ONE BLOCK

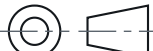
STEP	OPERATION	TOOLING	SHEET
1	Slit coil to 272.46 wide; decoil, level and feed	Low	H-2
2	Punch every hole, groove and notch in register	High	H-2, H-3
3	Roll form the profile; the hem closes in the last passes	High	H-1
4	Cut to length L after forming	Medium	H-2
5	Bend the four corners in a fixture: BR, TR, TL, BL	Medium	H-1
6	Butt weld the ends; grind flush; prime the weld	Medium	H-1
7	Projection weld the M8 nuts from inside	Low	H-0
8	Rivet the hinge pins into the shelf; clinch in the rails, shelf and strap	Low	H-0, H-5
9	Mask hinge pins, threads and joint holes; pretreat, powder coat matte black	Low	H-0
10	Pad print the gloss logo strip up both sides	Low	C-5
11	Fit the spring latch with its two screws	Low	H-5

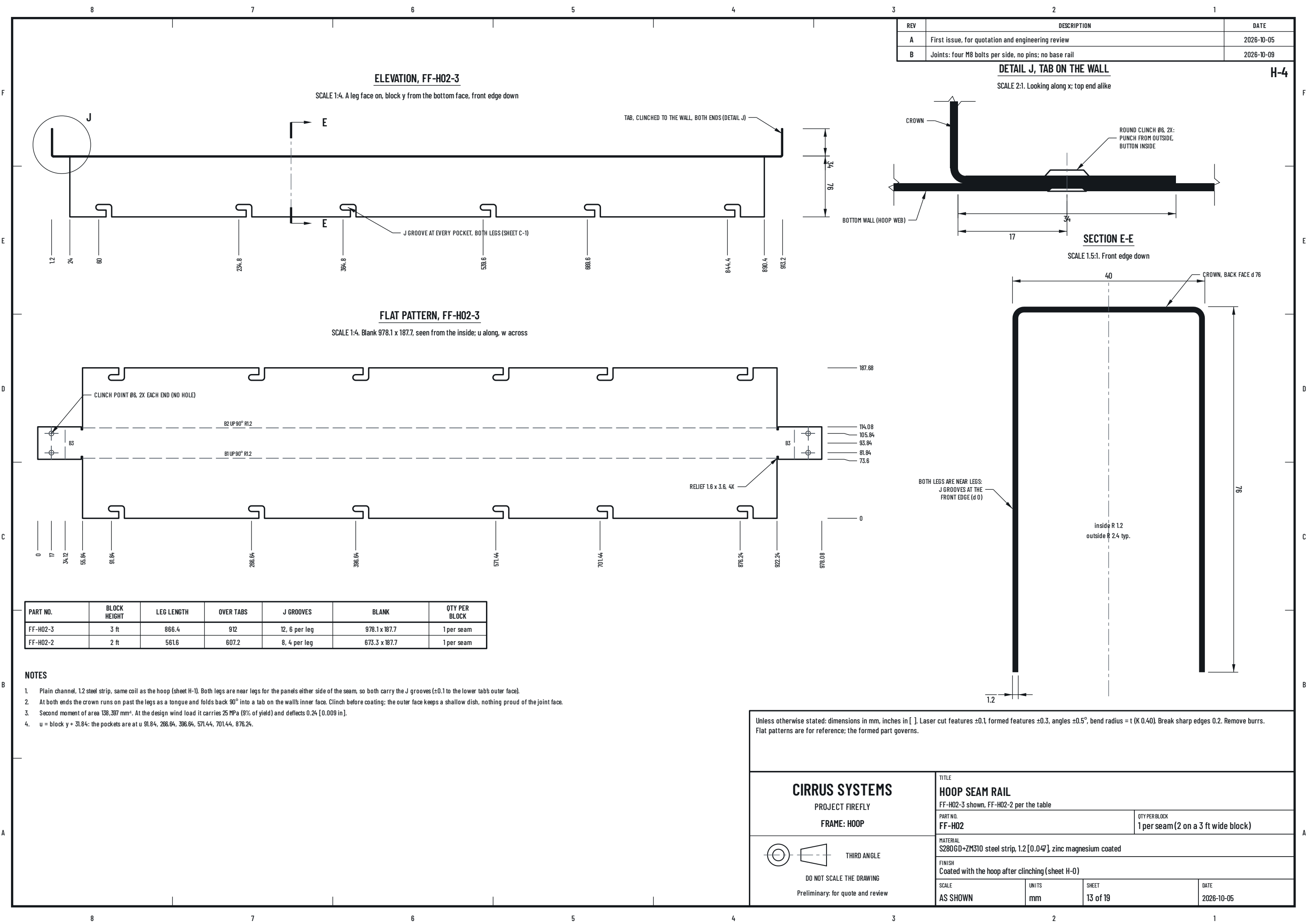
1. x runs from the block's left outer face seen from the front; y up from its bottom outer face. On a flat, s = the run's start + the distance from its corner (table above and view A). A count in { } is that many centers at the pitch, starting at the one given.
2. d is depth back from the frame's front edge, q is in from the outer face along the flange; w = d + 63.59 on the web and q + 213.82 on the flange. Positions ± 0.1 to the strip's punched datum, accumulated ± 0.3 along the full length.

NOTES

1. Strip: 272.46 +0.2/0 wide, slit from coil, Z1310 both sides, burr toward the inside face.
2. Punch every feature in register before roll forming; the J grooves, joint holes and corner notches share one datum per block length. Cut to length after forming.
3. The weld closes the ring at x254 on the bottom, between two flange holes and clear of vents (sheet H-1).
4. Logo strip band: up both sides the outer face from the front edge to the phantom line at w 95.59 (d 32) carries no cut feature, burr, tooling mark or scratch. The logos are printed after coating, sheet C-5.
5. Mass is the bare formed strip, before nuts and coating.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

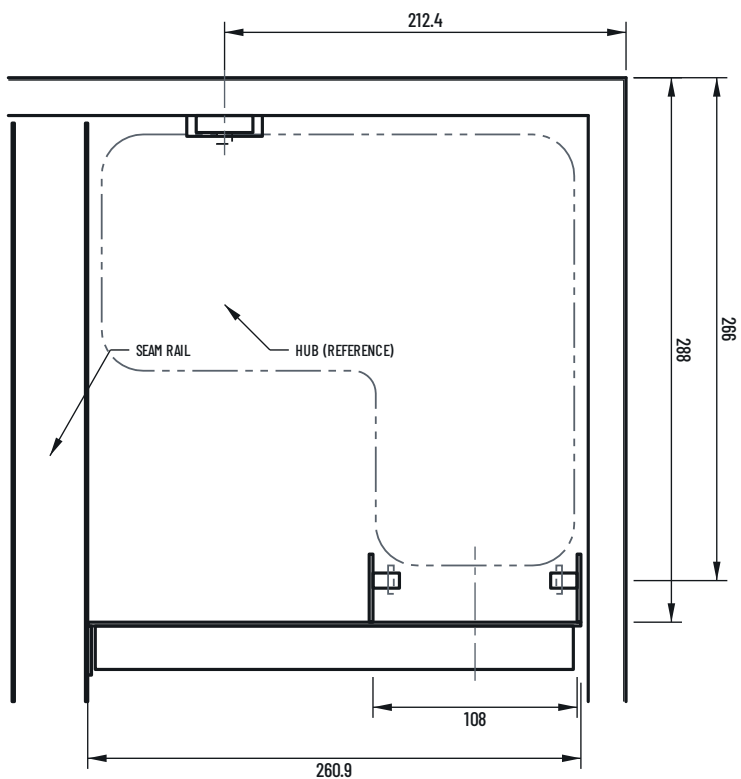
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE HOOP STRIP, FLAT PATTERN Strip lengths, corner positions and every cut feature for the four block sizes			
	PART NO. FF-H01-33 / -32 / -23 / -22		QTY PER BLOCK 1	
THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
	FINISH Bare at this stage: coated after welding (sheet H-1)			
	SCALE 1:10	UNITS mm	SHEET 11 of 19	DATE 2026-10-05



REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

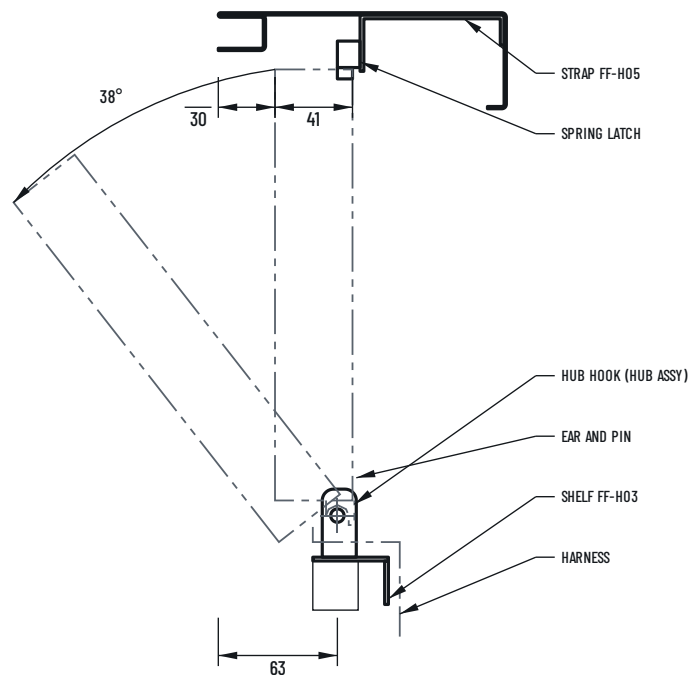
VIEW A, THE HUB BAY

SCALE 1:4. From the front, top right panel out



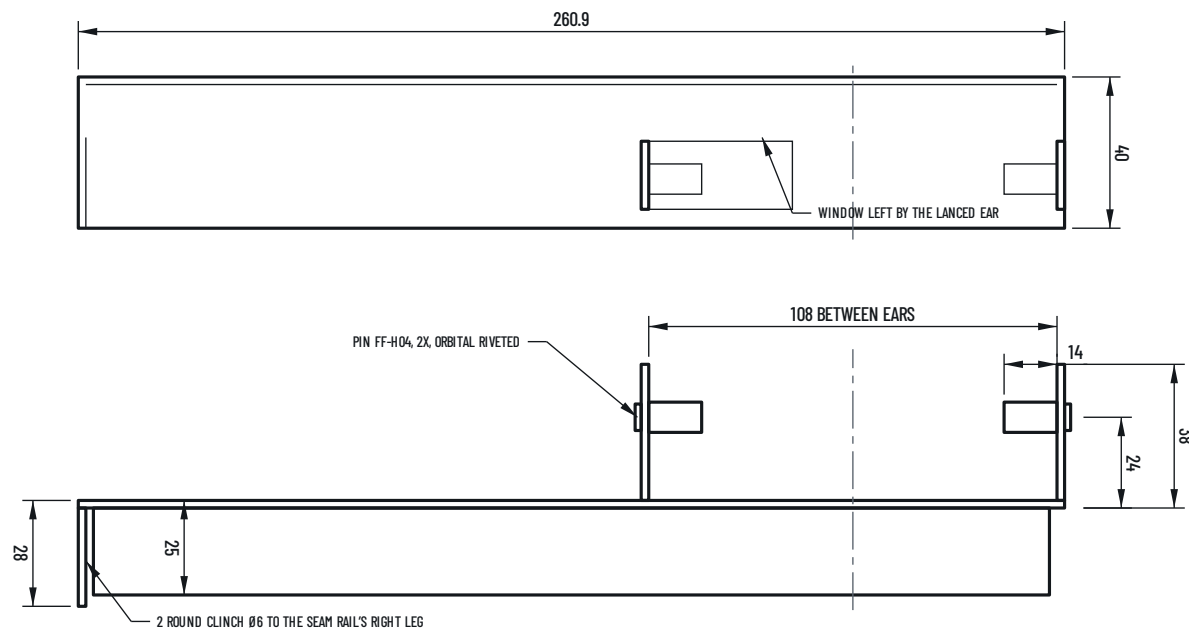
VIEW C, FROM THE RIGHT

SCALE 1:4. Side wall removed; hub upright and tilted



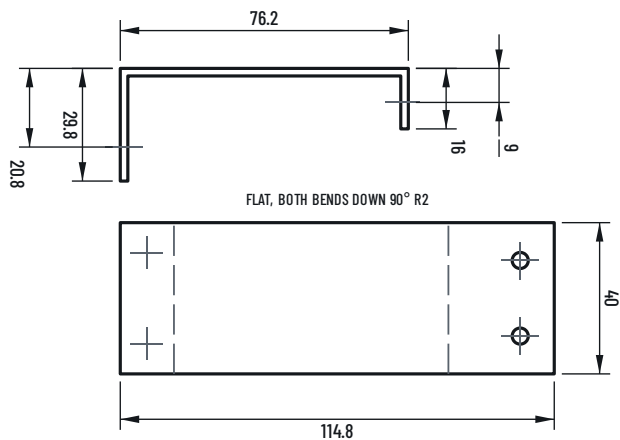
FF-H03 HINGE SHELF

SCALE 1:2. Top view and front view; x from the foot of the hub's leg



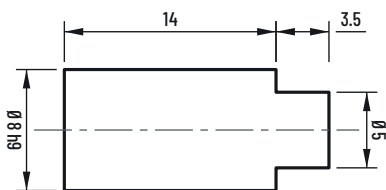
FF-H05 LATCH STRAP

SCALE 1:2. From the right; 40 wide



FF-H04 PIN

SCALE 2:1. 303 stainless

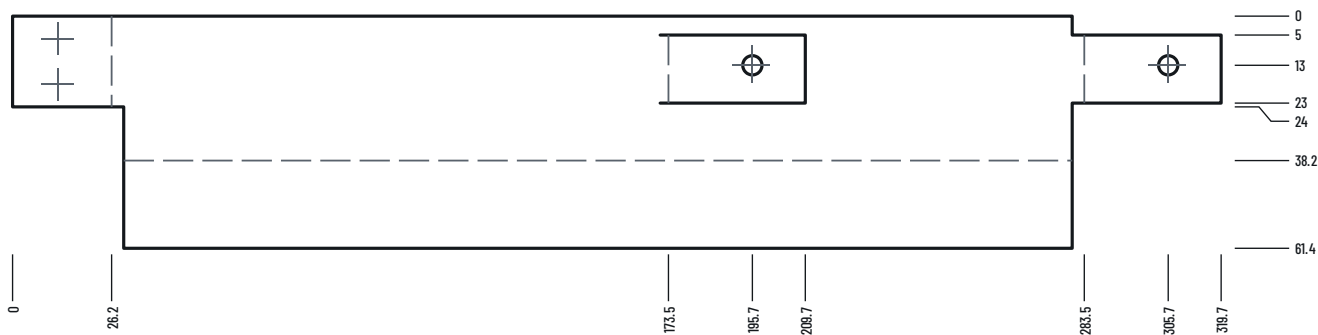


PART NO.	DESCRIPTION	MATERIAL	QTY	TOOLING
FF-H03	Hinge shelf with a lanced ear, clinched to the seam rail	Steel DX51D+Z275, 2.0	1	Low
FF-H04	Hinge pin Ø8, orbital riveted into the ears	303 stainless	2	Low
FF-H05	Latch strap, clinched to the top flange	Steel DX51D+Z275, 2.0	1	Low
CLS-M3-1	PEM self clinching nut M3, in FF-H05	Stainless	2	
LATCH	Spring bolt latch, push to release, 30 x 14 x 12	Purchased	1	
ISO 7045	Pan screw M3 x 16, latch to strap	A2 stainless	2	

1. Same parts and positions on every block size: the shelf hangs off the seam rail nearest the hub, so its length never changes. Nothing touches an outer wall.
2. Latch: bolt 8 x 8 with a 45° lead in, 6 throw, 10 to 15 N spring. It drops into a catch slot in the hub's top (hub assembly). The pins and hooks carry the hub; the latch only stops it tilting.
3. Rivet the pins into the shelf, then clinch the shelf and strap in before coating; mask the pins and the M3 threads. Hub swap: press the latch, tilt 38°, lift 12.7 off the pins, out the front.

FF-H03 FLAT PATTERN

SCALE 1:2. 319.7 x 61.4. Tab and back flange DOWN, ears UP, all 90° R2; Ø 5.1 pin holes

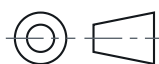


Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: HOOP



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

HOOP HUB MOUNT

Hinge shelf, pins, latch strap and latch; hub shown for reference

PART NO.

FF-H03 / FF-H04 / FF-H05

QTY PER BLOCK

1 set

MATERIAL

[See parts list](#)

FINISH
Shelf and strap coated with the frame; pins bare

SCALE

AS SHOWN

UNITS

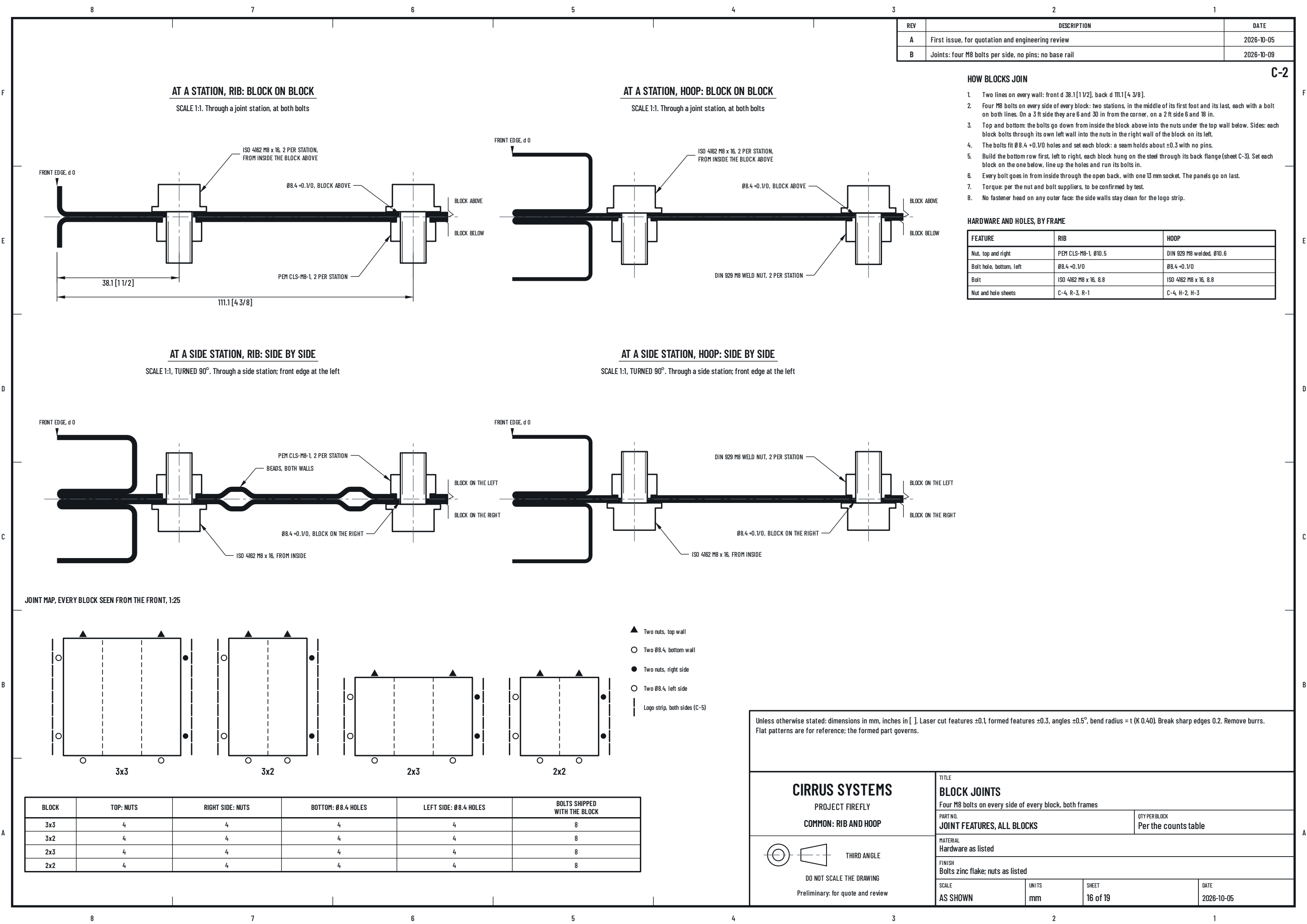
mm

SHEET

14 of 19

DATE _____

2026-10-05

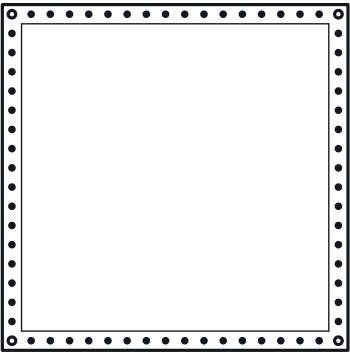


REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

C-3

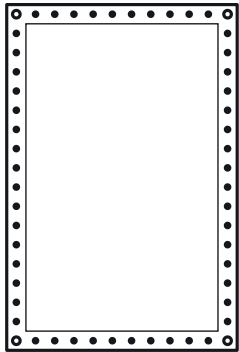
BACK FLANGE HOLES, EACH BLOCK FROM BEHIND

SCALE 1:20. Ø11.1 [7/16] holes, 25.4 [1] in from every block edge, every 50.8 [2] along it



3x3

RIB 68, HOOP 64



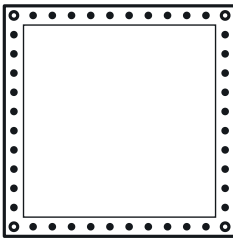
3x2

RIB 56, HOOP 52



2x3

RIB 56, HOOP 52

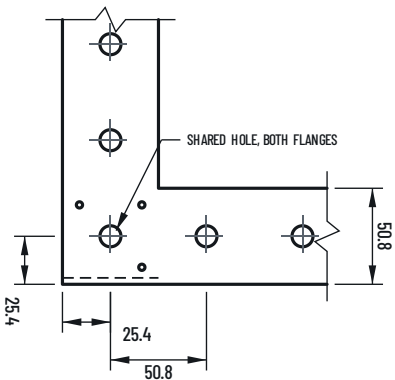


2x2

RIB 44, HOOP 40

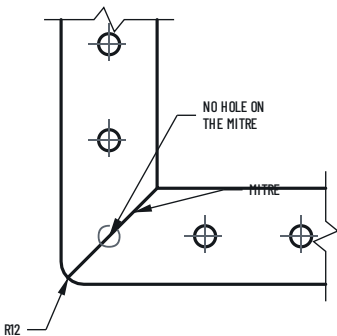
DETAIL K, RIB

SCALE 1:4. A back flange corner from behind



DETAIL K, HOOP

SCALE 1:4. The flanges meet on a mitre

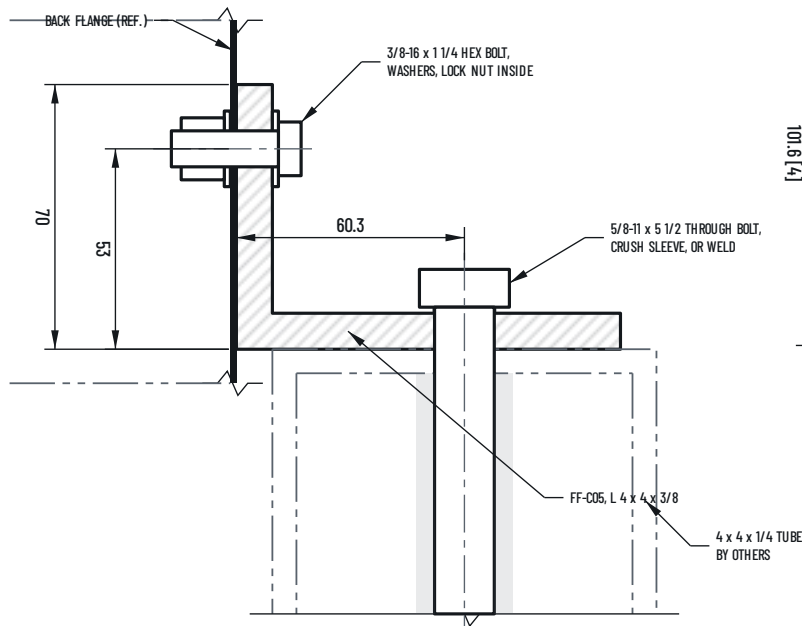


THE GRID

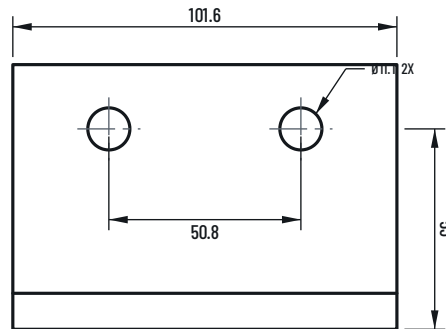
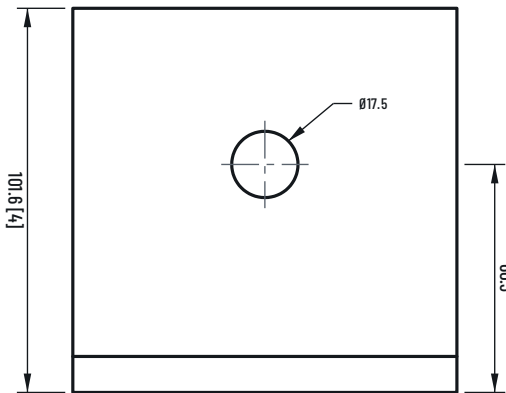
- Every block edge sits on a whole foot, so every hole on every wall falls on an odd inch both ways, and a hole line along a row or column seam runs unbroken across the wall.
- Angle: 2 x 2 x 3/16, by others or the angle kit, bolted with 3/8 in bolts every 152.4 [6] on the bottom hole line of every row and the top line of the wall. Rows are 2 or 3 ft, so angles stay under the 4 ft rule.
- Tube: at mid row on clips, one at each wall end and one per side seam.
- Every row hangs on the steel, the bottom row too, so nothing has to sit under the wall.
- Holes the steel leaves free take cable ties, P clips or conduit straps.

SECTION AT A CLIP

SCALE 1:2. Frame at the left, the tube behind it



FF-C05-2, FROM ABOVE



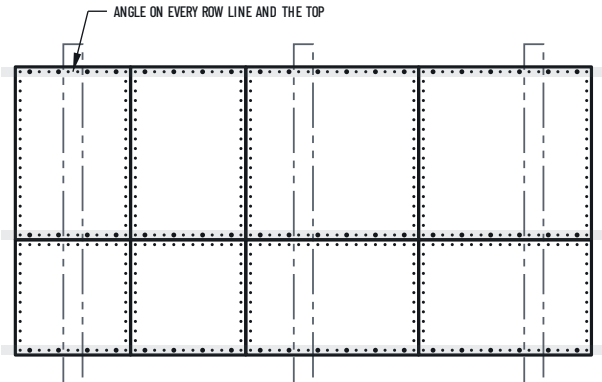
FROM BEHIND THE WALL

PART NO.	WIDTH	ØTL1 HOLES	USED
FF-C05-1	50.8	1, on the center	each wall end
FF-C05-2	101.6	2, at ±25.4	each side seam

- Cut from L 4 x 4 x 3/8 in steel angle (A36), vertical leg trimmed to 70. Hot dip galvanized after cutting and drilling (ASTM A123).
- A clip at a seam takes a hole in each block, so it also ties the seam.

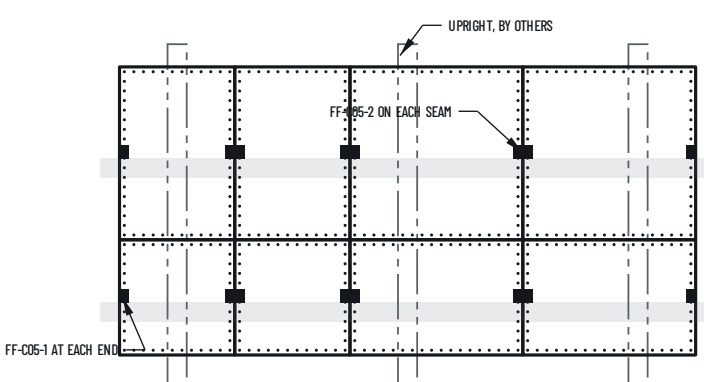
5 x 10 WALL ON ANGLE IRON

SCALE 1:40. From behind; uprights by others



5 x 10 WALL ON 4 IN TUBE

SCALE 1:40. From behind; uprights by others



STEEL PER PLANNER WALL (HEIGHT x WIDTH, FT)

WALL	BLOCKS	ANGLE LINES	3/8 BOLTS, ANGLE	TUBE LINES	FF-C05-1	FF-C05-2
2x6	2	2	24	1	2	1
3x6	2	2	24	1	2	1
3x8	3	2	32	1	2	2
4x8	6	3	48	2	4	4
4x10	8	3	60	2	4	6
5x10	8	3	60	2	4	6
6x12	8	3	72	2	4	6
8x16	18	4	128	3	6	15

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

COMMON: RIB AND HOOP



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

MOUNTING: HOLE GRID, ANGLE AND TUBE CLIPS

Back flange holes on one 2 in grid across every wall

PART NO.

FF-C05-1, FF-C05-2

QTY PER BLOCK

Per the steel table

MATERIAL

Clips: L 4 x 4 x 3/8 in steel angle, A36

FINISH

Clips hot dip galvanized, ASTM A123

SCALE

AS SHOWN

UNITS

mm

SHEET

17 of 19

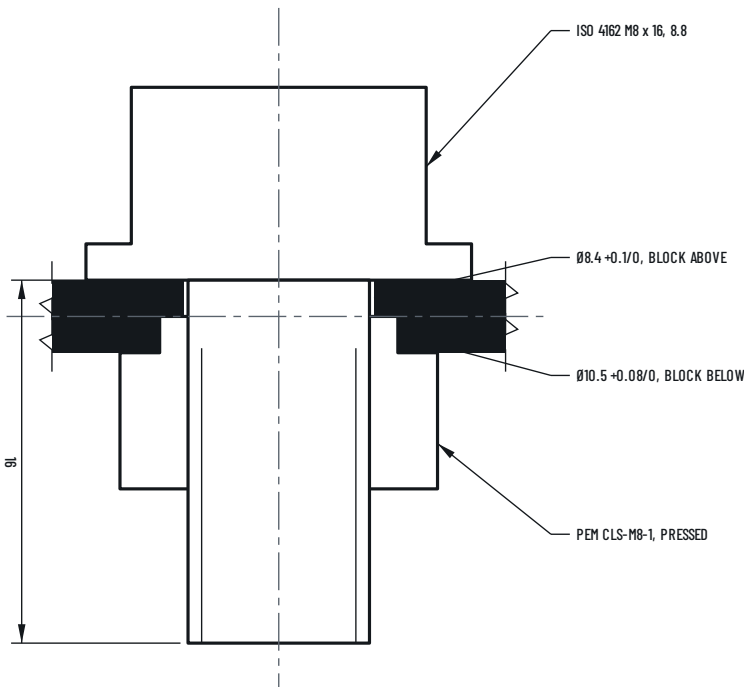
DATE

2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

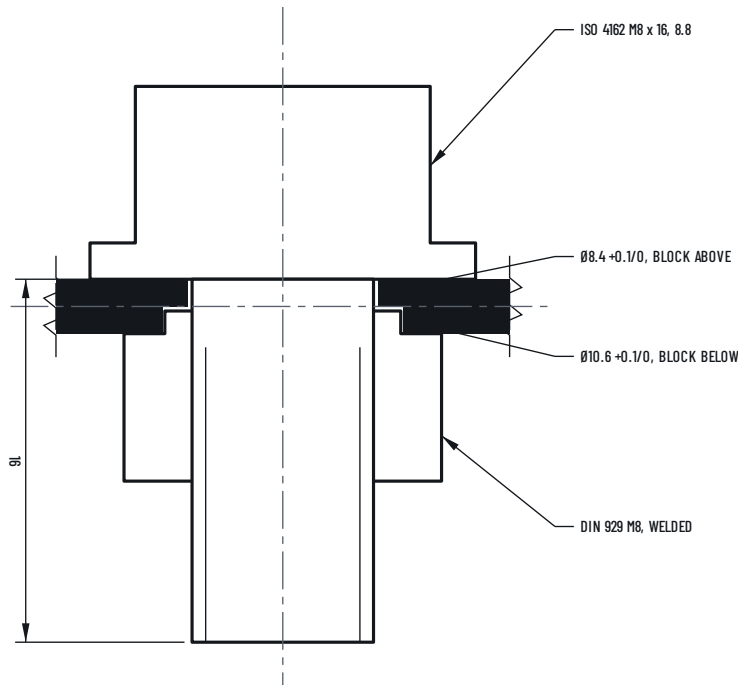
THE JOINT, RIB

SCALE 3:1. Block above on block below, 1.6 walls



THE JOINT, HOOP

SCALE 3:1. Block above on block below, 1.2 walls



THE JOINT

1. Every joint is one ISO 4162 M8 x 16 flange bolt, class 8.8, through a $\varnothing 8.4 \pm 0.1/0$ hole into a nut in the next block: a PEM CLS-M8-1 pressed into Rib, a DIN 929 weld nut on Hoop. Four on every side of every block (sheet C-2).
2. Shown block on block. Side by side the stack is the same, turned 90°: the bolt goes through a block's own left wall into the nut in the right wall of the block on its left.
3. The hole leaves about 0.2 a side around the bolt, and nuts and holes sit ± 0.05 to each other on a wall and ± 0.1 to the frame datums (sheets R-3, R-1, H-2), so the bolts set each block: a seam holds about ± 0.3 with no pins.
4. Rib: press the nuts from inside before forming, into $\varnothing 10.5 \pm 0.08/0$. Hoop: projection weld the nuts from inside after the ring is closed, through $\varnothing 10.6 \pm 0.1/0$ pilots, before powder coat.
5. Mask every nut thread and every $\varnothing 8.4$ hole for powder coat.
6. One T3 mm socket by hand does every joint, from inside through the open back. Torque per the nut and bolt suppliers, to be confirmed by test.

HARDWARE, PER BLOCK (EACH BLOCK SIZE, HEIGHT x WIDTH, FT)

PART	DESCRIPTION	MATERIAL	WHERE	3x3	3x2	2x3	2x2	3x3	3x2	2x3	2x2
ISO 4162	Hex flange bolt M8 x 16, class 8.8	Zinc flake	Every joint, ships loose	8	8	8	8	8	8	8	8
CLS-M8-1	PEM self clinching nut M8	303 stainless	Rib top and right side walls, 4 each	8	8	8	8				
DIN 929 M8	Hex weld nut M8	Steel, plain	Hoop top and right side walls, 4 each					8	8	8	8
RIV-32-CSK	Blind rivet Ø3.2, closed end, 100° head	Al body, A2 mandrel	Rib corners, tabs, rails, hanger	35	31	35	31				
CLS-M3-1	PEM self clinching nut M3	Stainless	Hoop latch strap					2	2	2	2
LATCH	Spring bolt latch, push to release	Purchased	Hoop hub lock (sheet H-5)					1	1	1	1
ISO 7045	Pan head screw M3 x 16	A2 stainless	Hoop latch to strap					2	2	2	2

HARDWARE, PER WALL

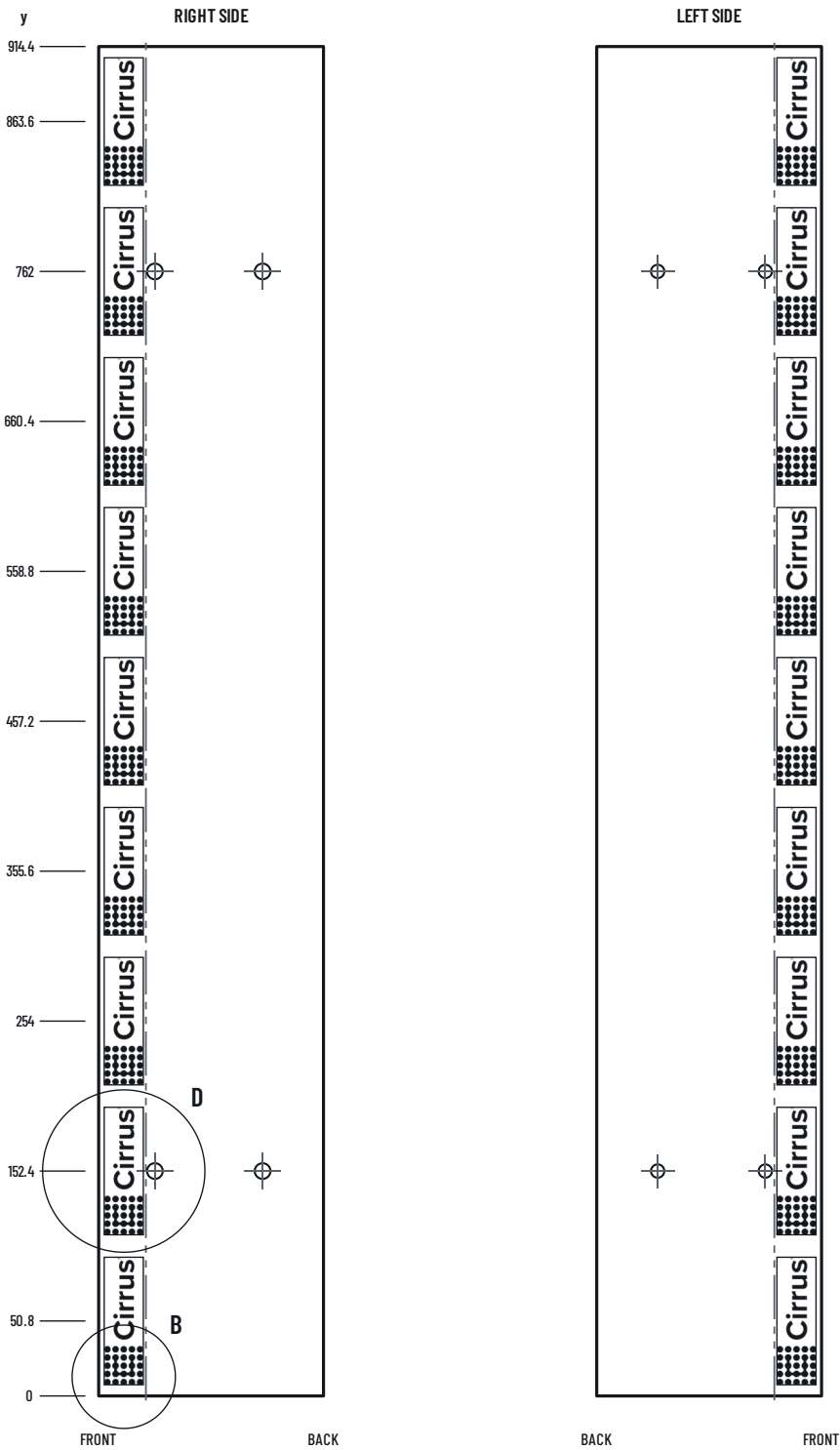
PART	DESCRIPTION	MATERIAL	QUANTITY	SHEET
FF-C05-1, -2	Tube clip, single and double	A36, galvanized	Per the steel table	C-3
CLIP BOLTS	3/8-16 x 1 1/4 hex bolt, 2 washers, nylon lock nut	Zinc plated, grade 5	1 per Ø1.1 clip hole	C-3
TUBE BOLTS	5/8-11 x 5 1/2 hex bolt, crush sleeve, washers, lock nut	Zinc plated, grade 5	1 per clip, or weld	C-3

WHAT SHIPS WITH WHAT

1. Each block ships with its eight joint bolts: one for every Ø8.4 hole in its bottom wall and its left side wall (seen from the front). They go into the nuts of the block below and the block on its left; a block on the bottom row or at the left end keeps the ones it does not use as spares.
2. Clips ship with their bolts. Angle iron and its bolts and the uprights are by others, or the optional angle kit. Every row hangs on the steel; nothing goes under the wall unless the installer wants it.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (X 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

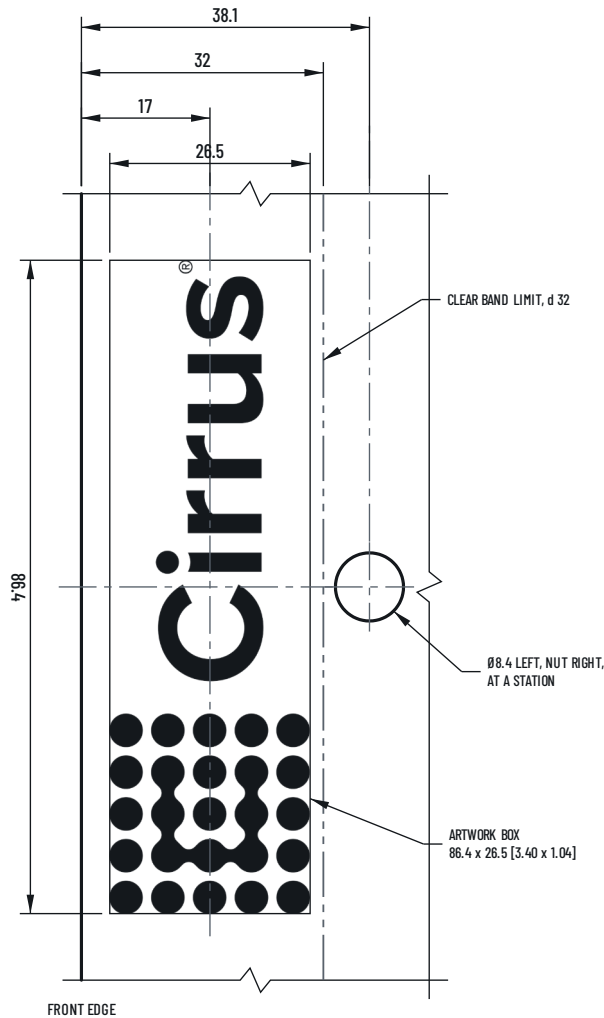
CIRRUS SYSTEMS PROJECT FIREFLY COMMON: RIB AND HOOP   THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE JOINT BOLTS AND HARDWARE The joint on both frames and every purchased part, per block and per wall PART NO. ISO 462, CLS-M8-1, DIN 929 QTY PER BLOCK Per the tables			
	MATERIAL As listed FINISH As listed SCALE AS SHOWN UNITS mm SHEET 18 of 19 DATE 2026-10-05			



Joint holes at the stations shown for reference. Vents, beads and the rest of each wall are on the frame sheets. Same pattern on a 2 ft wall.

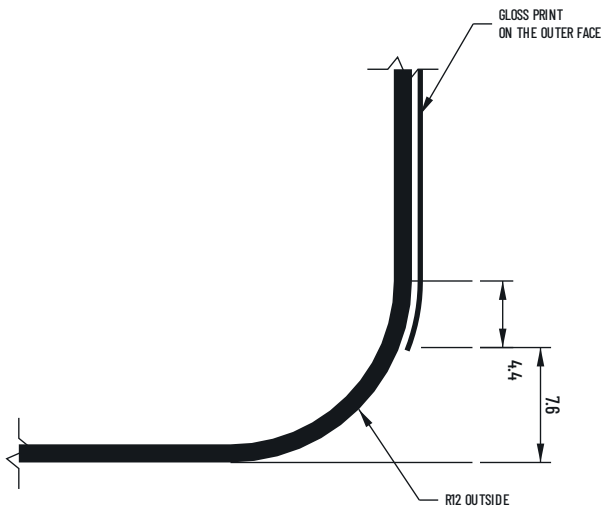
DETAIL D

SCALE 1:1. One logo at a joint station



DETAIL B, HOOP CORNER

SCALE 2:1. Cut at d 17, seen from the front



REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

C-5

LOGO POSITIONS ON EVERY BLOCK

BLOCK	SIDE WALL HEIGHT	LOGOS PER SIDE	LOGOS PER BLOCK	CENTERS, y UP FROM THE BLOCK'S BOTTOM FACE
3x3	914.4 [36]	9	18	50.8 + 101.6 n, n = 0 to 8 (to 863.6)
3x2	914.4 [36]	9	18	50.8 + 101.6 n, n = 0 to 8 (to 863.6)
2x3	609.6 [24]	6	12	50.8 + 101.6 n, n = 0 to 5 (to 558.8)
2x2	609.6 [24]	6	12	50.8 + 101.6 n, n = 0 to 5 (to 558.8)

ARTWORK AND PLACEMENT

ITEM	REQUIREMENT
Artwork	Cirrus wordmark, vector file supplied by Cirrus, printed at its own proportions
Size	86.4 x 26.5 [3.40 x 1.04], the wordmark's outer extents
Reads	Up the side on both hands: letter tops toward the front on the right side, toward the back on the left
Center	d 17 [0.67] back from the front edge; y per the table above
Tolerance	Each logo ± 0.5 in d and ± 1.0 in y to the block's bottom face, square to the front edge within 0.5°
Clear band	d 0 to 32 on both side walls: no hole, rivet, fastener, clinch, weld or tool mark

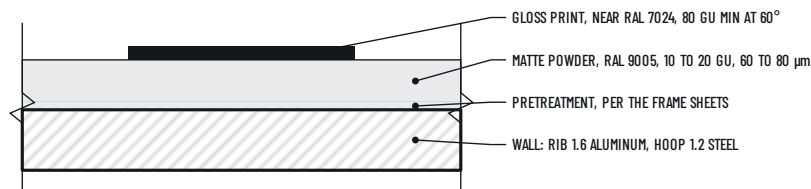
FINISH

ITEM	REQUIREMENT
Cabinet	Polyester powder, RAL 9005 black, matte: 10 to 20 GU at 60° (ISO 2813), 60 to 80 µm, over the frame's pretreatment
Logo	Outdoor two component pad printing ink, or UV cured inkjet: near RAL 7024 graphite gray, high gloss, 80 GU min at 60°
Look	Tone on tone, like a spot varnish: a dark gloss mark on the matte black, never white
When	After the powder is fully cured. Rib: side walls printed before riveting. Hoop: frames printed after assembly
Sample	Color, gloss and edge sharpness to a sample signed by Cirrus, printed on coated production metal
Adhesion	Cross cut, ISO 2409, class 0 or 1, on the print and on the coat
Outdoors	Supplier names the ink system and its outdoor rating; exposure test to be agreed with Cirrus

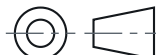
NOTES

1. Every frame, Rib and Hoop, is printed on both side walls, so any block can stand at either end of a screen. In a seam the two strips face each other; the print adds no measurable thickness.
2. The strip keeps its 101.6 [4] pitch across block seams: the top logo of a block sits 50.8 [2] below its top face and the next block's first logo 50.8 above it.
3. Hoop: the first and last logo on each side run about 4.4 onto the RTZ corner (detail B) and the pad conforms to it. Rib's corners are square, so its logos sit on the flat wall.
4. Keep the band free of scratches from coating to packing. Reject a print with voids, smears or a doubled edge visible from 1 m.
5. Joint holes at the stations sit level with a logo center: the hole edge is at least 2.5 clear of the artwork box.

PRINT OVER THE COAT, NOT TO SCALE



Unless otherwise stated: dimensions in mm, inches in []. y is up from the block's bottom outer face, d back from the side wall's front edge. Color and gloss to the sample signed by Cirrus.

CIRRUS SYSTEMS PROJECT FIREFLY COMMON: RIB AND HOOP	TITLE LOGO STRIP AND FINISH The Cirrus logo strip up both side walls of every block, Rib and Hoop		
	PART NO. SIDE WALLS, ALL BLOCKS		QTY PER BLOCK 18 logos (3 ft tall), 12 (2 ft tall)
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL Per the frame sheets		
	FINISH Matte RAL 9005 powder; gloss print near RAL 7024		
	SCALE AS SHOWN	UNITS mm	SHEET 19 of 19